

22AIE111 OBJECT ORIENTED PROGRAMMING IN JAVA

S2 B.Tech CSE-AI

Department of Computer Science and Engineering

Amrita School of Computing, Amritapuri Campus

CASE STUDY

PHASE – I (Analysis)

Problem Identification & Class Mapping (CO1)

Group No : 17

Team Members :

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1. Problem Title : Digital Complaint Management System

2. Problem Description :

Manual complaint systems are slow, lack transparency, and make tracking difficult. Users cannot easily monitor complaint status, and administrators struggle with assignment and resolution tracking.

This system provides a digital platform where:

- Users submit complaints
- Complaints are assigned to officers
- Progress is tracked
- Admin monitors the entire workflow

3. Identified Entities (Classes)

Class Name	Description
User	Client/ Customer using the application
Complaint	Represents a complaint with details and status
Officer	Handles and resolves complaints
Admin	Monitors system and manages complaints
System Manager	Core system managing complaint flow

4. Class Attributes and Behaviours

Class Name	Attributes	Methods(behaviours)
User	userId, u_name, u_email	submitComplaint(), viewComplaintStatus()
Attributes	complaintId, description, status, assignedOfficer	updateStatus(), assignOfficer()
Officer	officerId, o_name	resolveComplaint(), viewAssignedComplaints()
Admin	adminId, a_name	monitorComplaints(), assignComplaint()
System Manager	complaintList, users, officers	addComplaint(), trackStatus(), assignComplaint()